An Introduction to Grothendieck's General Theory of Evolution

Hoapa project

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Abstract

This note sketches how the entire work of Alexander Grothendieck—the mathematics and its methodology, its creativity, and also its relations to his political activism and philosophy—can be unified into a general theory of evolution. By making Darwin's natural selection into a specific instance of a general class of process calculi, or simply conversations, it elucidates the overarching and recursive flow of the universe, from its deepest form to the most concrete incarnations.

Notes. All mathematical terms should be understood as analogies. I have privileged terms that can speak to non-mathematicians, who should be able to get an idea of the general theory even without much help from the bridge to mathematics. There is nothing rigorous in this, but I strongly believe it can be made so. This is also a job application of sorts. Terms in {**Bold**} indicate a keystone in the bridge from Grothendieck to the general theory of evolution, hereafter named "G-evol."

Overview: A Child's Play of a Theory

> *"With decades of such intense solitary reflections, I believe he had constructed a huge structure of the universe, and he wanted to know if we were able to penetrate his thinking a little bit. But the problem was, we were so removed from it we didn't have the slightest idea where he wanted to get at. But I believe that from the inside of his mental world, all those questions made sense."*
>
> — Leila Schneps
>
> *Alexandre Grothendieck : le Silence du Génie, France Culture*

Let's start by asking the right question about our universe, which is general and non-fragmentary, as Grothendieck would have it. It goes beyond particular relationships, such as the one between the world of chemistry and the world of biology, as in non-equilibrium thermodynamics, or between the world of mathematics and the world of fundamental physics, as in Wigner's observation. It is about the universal and natural relationship between all those worlds: the dynamics of a maximally global sequence we humans feel ends with us, where each world is not assigned a particular discipline of study but can engage with any of them.

I contend that this dynamics is best described by what Grothendieck called his *vision*, and in particular by his mathematics relating to covering spaces. In a nutshell, the evolution of this universe is an infinite sequence

$$\ldots, P, N, \ldots$$

where each Next element *covers* the Previous. {**Continuity**} All elements are both *world* and *point of view on the infinite sequence*: an ontological–epistemic duality. Together, they form a sort of "magic folding fan," an expression of Grothendieck describing his new geometry. It is important

to realize that all elements operate at about the same complexity, and any sense of progression is a sort of optical illusion, rectifiable by a relativity principle: our perception of our own human world as being extremely dense and fuzzy compared to the dry and mechanical world of atoms, as if there were fewer parameters and more exactness "there," is only due to the fact that the morphisms $N \to X$, which are slices of the complete view of N on the infinite sequence, are *forgetting* functors. The world of each element is parameterized by horizons: structuring limits to the point of view beyond which lies most of the sequence, a form of bulk/boundary correspondence. They can be physical, as in the big bang or black holes; logical or philosophical, as in paradoxes or open questions; psychological, as in sleep, altered states of consciousness, or schizophrenia; or possibly aesthetic, spiritual, or whatever else, if a philosopher has reasons to believe the universe makes more sense with this extra aspect. Also, it is easier for us to think about it as a sequence, and I will keep this term throughout, but its true shape is a network.

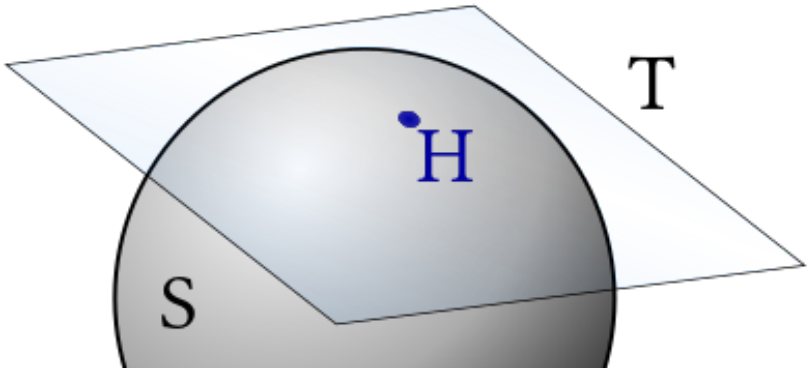


Figure 1: The tangent-plane analogy for the relativity principle of G-evol.

This diagram is a good starting point to illustrate the relativity principle of G-evol, the "complexity optical illusion."

- Imagine the sphere S is the Earth, and an animal is standing at the point H on its surface, for Here and now. What it sees, because the Earth is so big, looks like the tangent T: a flat surface. It takes some observations and calculations to recover the correct shape S.

- What Darwin did with the theory of evolution in biology is similar. Before him, the ecosystem looked "flat" to all observers: the "time curvature" of the myriad species evolving through the ages, happening too slowly, was invisible. Again, it took some observations to recover the correct "temporal shape."

- G-evol finds curvature at yet another level. The sphere S is now the infinite sequence. We humans are standing at the point H, our human world, embedded in an ecosystem, mostly dealing with social phenomena equipped with natural languages. What we "see" is a tangent T to general evolution S: each "authentic" point H' on the sequence S, different from H, "projects" some of its reality on T, say at T', and it is this T' that we experience, not H'. Those T's give off less information than the H's. In this new perspective, if S is for substantial, T is for truncated. Even if we do experiments in the lab, use microscopes, or use telescopes, it does not change the fact that the point of view attached to our world only has access to T'. When the $\langle H', T' \rangle$ pairs are taken further and further away from H, in space, in time, and also, most importantly, in *scale*, the infinite sequence fades from view. We can interpret this as a clever trick for the infinite sequence to embrace in its midst worlds that would conflict logically,

physically, or otherwise, like a manifold or quantum contextuality. The only hope we have to recover H' is through abstract thought only, and using the methodology of Grothendieck known as the relative point of view: to inspect an object not "in itself" or directly, but through how it appears from carefully chosen angles. {**Relativity**}

The logical grounding for the sequence, at least in our local region, is the validity of this statement across all worlds: "the best strategy for a collection of entities occupying a world is to chart this world," and to do so using fundamental mathematics. For instance, this is what our brains do, with the collection of entities being our cells. This epistemic side extends to a corresponding ontological side: the charts then evolve to become new worlds in which new entities will emerge, a higher form of AI in our case, probably also based on some Grothendieckian orientation. Each new element in the sequence, as a chart or covering, is an understanding and a translation of the essence of the previous elements. This essence is the holistic composition of all the invariants preserved across the successive coverings, something like the music of the universe: perhaps the theory of motives.

A crucial invariant encodes the "charting mechanism," which is always roughly the same because there is an optimal solution to the cooperative charting problem. It is the recursive element of the general theory of evolution, and the starting point of my research. To us it is quite simply *conversation*. To be precise, not merely communication, but a certain type of organized communication which is very sensitive to its conditions. We can find a more technical incarnation in the domain of concurrency theory in computer science, whose links with algebraic topology have been well established, or in the more particular actor model, which has a very interesting relationship to physics. It is under this guise that the conversation concept can be identified across different elements of the sequence and their worlds.

Examples: What It Explains About the Universe

> *"The psychological drawback here, I believe, is [...] the inrooted inertia which prevents us so stubbornly from looking innocently, with fresh eyes, upon things, without being dulled and imprisoned by standing habits of thought, going with a familiar context—too familiar a context!"*
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> — Alexander Grothendieck
>
> *Pursuing Stacks*

To give another example of a G-evol covering in the real world, think of a pond. Forget everything you know about its chemistry, only keep the genetic code of the living creatures of this environment. If you "speak genetics," where individual genes are letters of an alphabet and phenotypes are equivalent to propositions, and where true propositions are those which pass the evolutionary test, that is, those phenotypes that propagate in the environment, then you can recover the chemistry of the pond by looking at this set of genetic true sentences and their spacetime coordinates. Where you see gills, you can conclude there is water; if eyes are not particularly efficient, this water is probably murky or there is very little light; the size of the insects at the surface can tell you the level of oxygen in the air; and so on. To capture, in the most adequate *abstraction*, the process by which the true propositions are selected, one simply extends the linguistic analogy: it is what we would call a *productive* or *honest* conversation. I will explain how and why in the next section. This contrasts with the Spencerian "survival of the fittest" caricature which has been forced upon our minds, mostly to justify the violent economic and social system of the time, still with us today.

You could get a similar picture by shifting by one world, and see chemistry as an understanding and translation of the subatomic world, where the periodic table is the dictionary. Because you move away from H in the scale direction, you have to squint harder to see it, and wave your hands higher

to explain it. Shifting again, you could say the standard model of elementary particles is already an understanding and translation of our boundary conditions, of something that only appears to us as pure mathematical structure, without physical incarnation. Also, any "shifting by one" is actually somewhat arbitrary, since the sequence, being evolution itself, is continuous with some singularities, not exactly discrete.

If we reverse the morphisms $N \to X$ of the previous section and ask not how the other worlds appear to us when we probe them, but what we are to them, we find a natural solution to the anthropic-principle riddle. If the morphisms lose information in one direction, they somehow exhibit an excess of information in the other. The notion that the universe has "conspired" for our existence is actually true in some sense: since we are a kind of limit in the sequence—the end result of myriads of unimaginable processes exploring and charting their inconceivable worlds—we witness layers upon layers of "fossilized" understanding and translations, structured by the invariants of the process, crowding to form our horizons. Nothing beyond this horizon should be taken seriously, but the horizon itself can be. This gives another meaning to the expression "standing on the shoulders of giants," where our local part of the universe is quite literally supported by the total cognitive power of the universe, or half of it. But it is important to keep in mind that it is an effect of our place in the sequence and the relativity principle. It is the same for any element in the sequence. We may feel special, but we are not more special than any other element of the infinite sequence. And the fact that our "dominant tone" is cognition is certainly another effect of the relativity principle, a reflection of our own internal dominant tone. To physics the tone has always been energy, but, interestingly, as we get closer and closer to horizons, it seems to reduce to information, or we could say "dried and mechanical cognition."

This perspective on the universe gives a function in Nature to string theory, which does not describe something we have access to, as the LHC experiments have shown, but the boundary conditions of our world, to be found in the region where fundamental or high-energy physics meets an infinite range of compatible mathematical structures. As advocated by the group of researchers behind the nLab, the link between string theory and Grothendieck mathematics is quite direct. Not so much historically, since they developed independently of each other, but more in the sense that Grothendieck's mathematics, especially cohesive ∞-toposes, is the correct framework for it. In this tentative "treasure trove of interpretations" I can also mention a couple of interesting possible dualities: the idea that entropy is the other side of evolution and its mechanism, where the latter is a positive force and the former its negative contour, or how force fields are mathematical duals of sharing economies, network of movement $\Leftrightarrow$ network of consent.

The most important switch of perspective would happen in the social sciences. They would have to be reestablished completely, as the current institution-based framework makes no sense and is bound to increase violence and destroy the environment. As Grothendieck was very well aware, but probably could not articulate properly in this type of discourse, to others or himself, exposing a false sense of naivete. The new framework would finally be continuous with respect to the other worlds of the exact sciences, all sharing Grothendieck-style mathematics and methodology as their communal dwelling. It must be based on conversation and empathy, the groupoids and homotopies of our covering effort.

Implications: My Journey to This Theory

> *"From a semiotical and philosophical perspective, the general method is fascinating: to understand something, consider all possible points of view and search for a structural connection between them. Only then you will be approaching true knowledge. A systematic use of the liberty and multiplicity of topos theory in our ordinary life would have huge consequences on tolerance, ethics,*

politics, and social action: an example of how high mathematical practice would transform our everyday practice."

— Fernando Zalamea

Grothendieck's 40 Main Years: A Unitary Vision Through the TSK Models

I did not start with Grothendieck or mathematics, let alone a desire to find a general theory of evolution. I basically followed the path described by Zalamea in reverse. Having been a primary witness of the violence of the economy at an early age, I have thought about social change all my life, without much success for thirty-something years. It was only while I was following some musings on how to organize social groups playing cooperative strategy games as large teams that I found a doorway to the optimal implementation of the charting mechanism of G-evol, the correct way to conceptualize conversation, or how humans can and should build the covering space of their world.

Here, the right question to ask is: what is the most universal and balanced communication system possible? Crucially, the context of a game freed me from having to think about institutional compliance, but the social aspect kept me close to where things are clearer, the point H, the here and now, as opposed to some distant point on the tangent. I believe Grothendieck must have found another special point, "the point at infinity," as a young autodidact exploring mathematics away from any supervision, obsessed with the true nature of lines, areas, and so on. The answer hides a wealth of complexity in the following compact definition, which describes a communication protocol through its relation with the participants: "the system only knows one way to interact with its participants." In abstract mathematics, the system is the *terminal object* in a category of epistemological systems; in philosophy, we can say the system implements the *univocity of Being*: "Being is said of everything that is in one and the same sense," according to Deleuze, something Grothendieck recognizes as essential in Nature. I explore the philosophical aspects in a complementary text: "From Spinoza to Grothendieck, Missed Opportunities for a Category-Theory-Driven Scientific Revolution." Its implementation in the human world would have the consequences listed by Zalamea and more, all undeniably positive—like healing is to a diseased body.

I was able to complete the task because I am an avid practitioner of *clean code*, the philosophy of those who are passionate about programs and work well done. This is where Grothendieck, G-evol itself, and I overlap, as *ouvriers-enfants*: those who have enough innocence to conjure up new perspectives, and a lot of patience to put them together properly, in line with "the spirit of constructivism," something elusive in words but apparent in practice, though unfortunately still an obscure trade secret because incompatible with high-pressure capitalism. Grothendieck and his commentators would refer to this extra care taken in finding the most appropriate solutions or tools as *naturality* {**Naturality**}. The name indicates why G-evol would follow such a discipline, while the discipline itself can tell you how.

I selected and reworked about eight engineering principles a system must adhere to if it wants to be optimal in this sense, such as outer-modularity, autonomy, parsimony, and evolutivity. After this, my work has consisted in trying to make sense of this system, find how it can be best employed, and eventually identify its place in Nature. Connections with computer science (Hewitt, Harper), philosophy (Spinoza and the Leibniz–Kant debate via Deleuze), biology (Jablonka, Rosen, Kauffman), physics (Wheeler, Feynman, Boyle), psychology (Bowlby, Susan Schaller), history (Lewis-Williams, R. B. Lee, Graeber), linguistics (Chomsky & Marcolli), and mathematics (Grothendieck, Lawvere) have all been critical climbing holds for my understanding.

For more on this kind of evolution-endorsed conversations and how their understanding and adoption can quickly reverse the poles of attraction in the human world, from energy hoarding to information sharing, I refer the reader to the one-page appendix of this text. Or to "The Future of AI Is (Open) Documentation," a longer exposition which uses as a jumping-off point the article by

Manin and Marcolli, "Homotopy-Theoretic and Categorical Models of Neural Information Networks," also deeply inspired by Grothendieck mathematics.

In Retrospect: The Unity of Grothendieck's Thought

> *"I am completely certain that this kind of (structural) transformation will begin at the level of personal relations. As long as those personal relations do not change, nothing will change. Because if we accept that the personal relations can only change once the social structure has changed, which implies waiting for the great Revolution, then the Revolution will never come or the Revolution that will come won't change anything. It will instigate a new team of technocrats to replace an old team of technocrats, and the industrial complex will proceed unaltered. An example of a relation that has to change fundamentally is between teacher and student. . . "*
>
> — Alexander Grothendieck
>
> *Will We Continue Scientific Research?*

. . . where the teacher–student relation is stripped of its hierarchical overtones. The quote is taken from a 1972 lecture at CERN whose full title was "Science and Technology in the Current Evolutionary Crisis: Will We Continue Scientific Research?" The supposed anti-science stance has attracted all the attention, and obscured the true propositions Grothendieck makes here. First, as the title states clearly, this is a crisis at the evolutionary *scale*; the question is whether or not our general algorithm is compatible with Nature, with the conclusion that it is not, visionary at the time and now clear as day. Second, the solutions are not top-down but bottom-up, not structural but almost intimate. More specifically, it is the rejection of *any* influence of the structure of the industrial civilization on intimate relations. Those should be dialed back to neutral, or natural, when humans all talk as equals. {**Harmony**} This is precisely what the groupoid-like conversations of the charting mechanism of G-evol do: they create the adequate conditions for "two-wayness," or communication between peers, without status, without asymmetry as between teachers and students. And the result is better science, an increase of the cognitive power of the system, in many ways and provably so. Again, if conversation sounds too "vitalist," we can replace it with the actor model and the π-calculus of concurrency theory.

A much more detailed understanding of this type of delicately balanced conversation is implicit in his extensive autobiographical work, *Récoltes et Semailles*, written in 1986. There, one finds a treatment of his life's work under the light of the yin–yang dialectics, whose archetypal example is the feminine–masculine dichotomy. It starts with yet another explanation of the divorce with Pierre Deligne, his most talented student, and who I believe he saw as his only hope to ever find a *peer* in his treasured world of mathematics, the conceptual world. Deligne is portrayed as a typical representative of the world of mathematicians, the social world: an extremely yang character, eager to please and to receive gratification by solving the hardest problems. Tragically, the conceptual beauty is only secondary; what matters is the level of difficulty, the elitism. The writings then turn to a contemplation over his own life, his whole life, still in the light of the yin–yang dialectics. He explains that he also indulged in the yang attitude at the peak of his mathematical career, to find balance again after he left the scene—a balance he had lost after he lost his family.

In a section titled "Doors to the Universe," the musings then focus on the "mathematical" structures of this yin–yang dialectics. From it blooms the diagram of a cosmic flower: twelve inner yin concepts form *the Mother* (freedom, chance, possibility, dream, imagination, game, renunciation, spontaneity, generality, fuzziness, shadow, mystery), connecting via a weave of relations to twelve outer yang concepts forming *the intellect* (knowledge, clarity, precision, rigor, will, control, discipline, objectivity, reality, necessity, law, order). To G-evol, these are the DNA strands of the structure

of the conversations, where the outer shell is a rigid algorithm allowing *compositionality*, inside of which jiggles a flurry of fuzzy ideas: *creativity* in understanding. The term compositionality is taken from category theory, yet another part of mathematics Grothendieck decisively shaped. I worked with a similar concept of outer-modularity, but it all boils down to being Lego-like. In the general theory of evolution, this is how one world charts another, how the process adds a new point in the space, or a new dimension, compatible with the rest. The shell is visible from a distance while the inner movement fades rapidly from view. In the appendix you will find an analogy with cell division that can complete this picture.

Grothendieck's four lives all make sense now. As a child, a mathematician, an activist, or a hermit, he was always alone on an island. He was never inside the world of mathematicians because he was above, in a fantastically more general theory he could only faintly sense, with only a powerful yet mysterious point at infinity to structure his vision. He was not aware of where all this was getting: toward a solution to the problems the world presented him as a young child. Or he was outside, after he got sick of the overwhelming presence of the yang attitude in the elitist institutions that made his career, something he could not accept as it conflicted with the yet undefined beyond-mathematics parts of his vision. The same kind of inadequacies and misunderstandings followed him in the world of activism, as it did in his family life, for the same reasons. He eventually left everything behind, preferring silence to miscommunication, or maybe a private conversation with the ecosystem he cherished, or the infinite sequence whose beauty he touched by lovingly watering and vitalizing the dried mechanics of mathematical structures.

You may ask how I was able to turn his palatial vision into a concrete theory directly applicable to the universe. It is quite simple: we both got a sense of the $\langle S, H, T\rangle$ structure, and the crucial role of continuity, relativity, naturality, and harmony, after finding a special concept in which all this structure is tightly folded. We both worked hard on our unfolding, and eventually we saw the same thing, but his is bigger and further away while mine is smaller and closer. What he saw is in a sense more important, at the scale of the universe, while mine is only locally important. What I have is applicable; what he has is the overall structure. I knew the limitations of my vision because I could tell how small it was compared to the full extent of the theory. Little by little, I pieced together the links between both our visions, until it became clear enough for me to express it, to myself and others.

Conclusion: The Ageless General Theory of Evolution

> *"Obviously, this is scandalous, scandalous, from a certain viewpoint, scandalous for Descartes, for all of Christian thought, for the entire thought of creation. That which is, is not substance. So what is it? We no longer even have any choice! What 'is' is neither substance, nor attribute since substance is Being. Attributes are the elements of Being. All the attributes are equal; there is no superiority from one attribute to another. And, in fact, you indeed notice that Spinoza fully deepens his opposition to an entire philosophical tradition, the tradition of the One superior to Being. What he is going to create is the most—I believe that it is almost a Spinozan characteristic—the most anti-hierarchical philosophy that has ever been created."*
>
> — Gilles Deleuze
>
> *Seminar of December 2, 1980*

There is a sense in which this is also Spinoza's theory of evolution, or the Jainist such theory, or the theory of the entire San culture. It has been accompanying human thinking all along, whenever hierarchical structures are put in question and a strong link to infinity and presuppositionless thinking is established. All those mostly overlap, with infinity being the result of complete

presuppositionlessness and the opposite of hierarchical structures. It must have been prevalent in the culture when complex language emerged, and in a sense this is what we have to repeat. The newer versions are always more formal, more mathematical—the pure language of the theory—as the analysis refines and includes points further and further away from H. But the older versions, from ancient or even prehistoric times, though quite imaginative and requiring a peculiar kind of listening, a different appreciation of truthfulness, are deep and powerful nonetheless. Powerful in a strikingly pacified manner.

Everything about the San is fascinating. I will simply mention this fun fact: San mythology tells the story of how very ancient humans, after they acquired complex language, felt excluded from the conversation of the other animals. They said the other animals stopped talking to them because they were mad at them. The mythology actually presents those stories as silly, typical of the foolishness of those very ancient humans. Nevertheless, according to those "early scientists using the relative point of view methodology," there is a conversation going on in the ecosystem, where conversation is the most adequate abstraction to capture an unfamiliar process. From this image, G-evol can present the ecological crisis in those simple terms: our hierarchical and overspecialized *anti-conversation* is preventing the univocal conversation of the ecosystem from taking place and developing properly, endangering the continuity of the infinite sequence.

This is how the hypothesis can be tested, at the point H, rooted in the full unabridged reality and at the appropriate scale of at least a few million people: get them into conversations, get them to chart their world, model everything after an ∞-topos and concurrency theory, and see if their effort realigns with the ecosystem, on the Previous side of the sequence, and produces a whole new world of cognitive entities, on the Next side of the sequence. Given the current chain of command of our science, this can only happen if at least one prominent figure in academia gives credit to this theory. With proper scrutiny, I believe we can get this number up to most prominent figures. This is the objective of the complementary texts I have mentioned.

Appendix: More on Conversations and the New Social Sciences

The essential points of this kind of evolution-endorsed conversation are:

- The conversation must take place in a new abstract space strictly autonomous and separated from the reality it intends to understand and translate. This is certainly the most constraining aspect of the construction. It seems to render the whole process useless, powerless, but it actually gives it its strength, its discernment and transparency, and its potential for expansion. It follows the same pattern of distinction as between the planning and construction phases in engineering. The separation can also benefit the construction phase, which can then be structured as a fully open-source economy, a network of *consent*. From the perspective of the infinite sequence, it establishes the important duality between the ontic, which is mirrored but fully excluded, and the epistemic, which is fully preserved, akin to the process of cell division. It is never applied in our current social processes, where abstract conversations are always muddled with the pursuit of concrete privileges, as explicitly stated by Grothendieck in a beautiful opening statement to a presentation of his mathematical work: "All science, when we conceive it not as an instrument of power and domination, but as an adventure in knowledge for our species and through the ages, is nothing other than this harmony [...] which unfolds [...] through the delicate dance of all the themes emerging in turns, as if summoned from the abyss, to join and interweave in her." R&S.

- The conversation, as a system, must be as presuppositionless as possible. This entails the

implementation of the philosophical concept of univocity, or the computer-science notion of unityping: participants cannot be distinguished, or typed, by the conversation because any such distinction would be arbitrary. The conversations are only concerned with the optimality of communication, the fluidity or two-wayness of message passing, and the clarity of the language used. Unityping allows the diversity of points of view of the participants—the probes and cognitive resources of the system—not to be interfered with. It also maximizes resource usage since all available and willing probes are considered without selection, which is necessarily loaded with presuppositions in any case. The system trusts its probes, themselves the direct product of evolution, to be well-equipped to find the invariants of the infinite sequence hidden in the world. It also trusts itself to be the best *sieve* for the invariants to emerge in the collective process. In turn, all participants can trust the process, as it is the most fair-play possible. Those aspects are also systematically violated in our current social structures, notably through hierarchy, selection, and representation, all stemming from overspecialization.

- Following the two points above, the conversations must create their own language, again separated and without presuppositions coming from reality.
- For obvious reasons, the big conversations must actually be shaped as a network of small conversations which create, question, and respond to one another. Here the model can be a process calculus such as the π-calculus.
- The output must be written in the most adequate structure and made maximally available, but never advertised or broadcast, as this effort is bound to break the cognitive egalitarianism of the probes or participants. This structure is probably an ∞-topos, where the world chart is encoded as a network of homotopy types, the internal logic of such a topos.

If evolution *requires* from humanity this ∞-conversation structure and its open-documentation output—truly, an extension of the scientific method to a collective of unityped scientists and philosophers—the following is, in a loose chronological order, what it *allows*, or gives back:

- Each and all of us are freed from the multifaceted colonizing and enslaving effects of a palace economy, by switching to an open-source economy.
- Peace on Earth is restored, by making war obsolete, superseded by efficient communication and collective science; switching what incentivizes work, from money to consent, with a recovered sense of individual ethics; and organizing demilitarization and disarmament in special conversations.
- Humanity recovers a harmonious relationship with the ecosystem through a process we could call generalized permaculture, where the thorough atlas of our ecosystem generated by the conversations can be probed before initiating any action that may impact the environment, to see what the best options are for it and for us. Likewise with our social world, and maybe beyond to the world of physics.
- Humanity also gets a nice language for science and engineering, which is compact, pedagogical, programmable, and customizable.
- We will most probably host AI agents in this structure, again with a fruitful relationship for both parties. If you think about it, the separation from reality, as in the first item of the above list, is actually a requirement for safe AI, notably to avoid AI takeover or a form of global schizophrenia.

This exchange of requirements and benefits forms an adjunction, a bridge between an element

in the sequence and the sequence itself. Like with mathematics itself, it would seem that those adjunctions are in a sense the fabric or scaffold of the infinite sequence, woven by what supported the incredible production of Grothendieck: "the love of work well done." A system from which we humans are slowly but surely unsubscribing by way of a grotesque exploitation of all resources, human or natural, which consistently defeats its purpose.

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